

Morgan Kim

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EDUCATION

Clearwater Institute of Technology | B.S. in Electrical Engineering | Metro City, USA | Sep 2017 - May 2021

EXPERIENCE

Voltline Robotics | Embedded Software Engineer | Metro City, USA | Mar 2023 - Jun 2025

- Rewrote the motor-controller firmware for a warehouse robot fleet of 300 units, cutting control-loop jitter from 120 to 15 microseconds and field motor faults by 60%.
- Cut firmware update time over the air from 11 minutes to 4 by compressing images and resuming interrupted downloads.
- Wrote a hardware-in-the-loop test rig that ran 400 motor-control tests on every firmware build.
- Added a crash logger that saved the last 2 seconds of sensor data to flash, which found the root cause of 5 field failures.
- Presented the firmware roadmap to the hardware and product teams each quarter, and two board revisions followed its recommendations.
- Documented the bootloader and flashing process, so new engineers could flash a board on their first day.

Clearwater Sensors | Hardware Test Engineer | Metro City, USA | Jul 2021 - Feb 2023

- Automated end-of-line tests for a humidity sensor, cutting test time per unit from 90 to 25 seconds across 20,000 units a month.
- Traced a batch of early field failures to a solder-paste change; the findings updated the reflow profile and cut returns from 2.1% to 0.4%.
- Wrote the calibration procedure for the sensor line, cutting calibration drift complaints from 15 to 2 a quarter.
- Designed a test fixture that checked 8 boards at once, raising station throughput from 40 to 150 boards an hour.

PROJECTS

Low-Power Weather Station | Independent Project | C, STM32 | Jan 2025 - Present

- Built a solar-powered weather station on an STM32 that reports every 5 minutes over LoRa and runs 9 days without sun.
- Cut average current from 4.2 mA to 0.9 mA by sleeping between readings and batching radio transmissions.
- Published the schematics and firmware openly, and a local school built two stations from them.

Formula Student Electric Car | Electronics Lead, Team of 12 | C | Sep 2019 - May 2021

- Designed the battery-monitoring board for the team's electric car, which passed technical inspection on the first attempt.
- Wrote the CAN-bus firmware for the dashboard and tested it against the motor controller before each race.
- Mentored 4 first-year members on soldering and board bring-up.

SKILLS

Tools: C, C++, STM32, oscilloscopes

Methods: firmware, hardware-in-the-loop testing, CAN, low-power design